Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

Reviewed example = a checked experiment; GenAge annotation = catalogue evidence only.

All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C110 **pme-1** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 pme-1** GenAge 1757 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 23870130	No gene in this candidate network	**M1 Parp1** GenAge 1010 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 20561897
Direct pairs: W1–M1 (1:1)				
C093 **pnc-1** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 pnc-1** GenAge 741 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 17335870	**F1 Naam** GenAge 871 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 18678867	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				
C115 **PTC5** 2 genes / 2 species 1 direct pair	**Y1 PTC5** GenAge 1373 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20883526	**W1 pdp-1** GenAge 2065 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 21533078	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C102 **Pten** 2 genes / 2 species 1 direct pair	No gene in this candidate network	No gene in this candidate network	**F1 Pten** GenAge 876 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 21111239	**M1 Pten** GenAge 1012 **GenAge annotation** **Pro-Longevity** Lifespan: Increase + Decrease Refs: 17400585, 22405073
Direct pairs: F1–M1 (1:1)				
C066 **RPD3** 2 genes / 2 species 1 direct pair	**Y1 RPD3** GenAge 555 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 10512855	No gene in this candidate network	**F1 Rpd3** GenAge 556 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 12459580	No gene in this candidate network
Direct pairs: F1–Y1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.